Services, technology and standards for broadcast-related multimedia

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This paper summarises the different broadcast TV delivery options in terms of current analogue systems and the progress on development of digital TV standards world-wide. It then describes the specifications that are being developed by the European Digital Video Broadcasting (DVB) project to enable delivery of new digital broadcast TV services in a uniform manner over all practical delivery platforms to low-cost set-top boxes. The requirements and the system implications for addition of PSTN- or ISDN-based interactivity are then briefly discussed.

1. Introduction

Broadcast television is perhaps the most widely accepted form of multimedia communication today. Since its introduction it has had an impact on social, economic and political development world-wide at least comparable with that of the telephone. The more recent introduction of practical video recorder/players has added the capability to view broadcast services at a more convenient time and to view non-broadcast material which is purchased or rented from a wide range of outlets on impulse. This complementary facility has been widely adopted by consumers, as has the remote control. Teletext was not, however, taken up quite so enthusiastically and Prestel was even less well received which shows that it is important to ensure that consumers will find a new offering attractive before committing to it.

Broadcast technology is developing rapidly and the change from analogue to digital TV delivery is almost upon us. The main advantage for conventional broadcasters will be increased capacity and flexibility and this will be used to increase the range of programming to cover a wider range of information, entertainment and transactional services and to offer increased control through ‘staggercasting’ which is a system where material is transmitted many times with relatively short time delays between each broadcast. There are clear plans to use staggercasting for popular movies and this is referred to as near video on demand (NVoD), but the same techniques can also be applied to news and other material.

It remains to be seen, however, how much extra money consumers will be prepared to pay for these new services and for the equipment to receive and decode them. There is therefore considerable interest in adding an extra dimension to the new services by enabling network interactions to be associated with broadcast services in a practical manner. This would combine the functionality of the existing telephone network with that of the TV set to allow audio, text and still-picture information to be provided on demand.

This could provide a practical and attractive stepping stone towards other new network services such as videophone and video-on-demand by building up consumer and information-provider awareness of the ways in which a TV display and a remote control can be used for personalised services. It will, however, be necessary to learn from the experiences with Prestel and ensure that the services are appropriate for a TV environment rather than a personal computer environment.

This paper mainly concentrates on the specifications which are being developed by the European Digital Video Broadcasting (DVB) project to enable delivery of new digital broadcast TV services (including interactivity as an option) in a uniform manner over all practical delivery platforms to low-cost set-top boxes. The benefits from the resulting mass market will enable manufacturers to provide a range of interoperable equipment (at least for satellite and digital terrestrial TV) for sale in the high street.

Sections 2 and 3 set the scene by summarising the different delivery options in terms of current analogue systems and the progress on development of digital TV standards world-wide. Section 4 presents the main achievements of the DVB project in greater detail and section 5 discusses the requirements and the system implications for addition of PSTN- or ISDN-based interactivity.

2. Analogue TV broadcasting

Broadcast TV is normally formatted using PAL or Secam in 625 line/50 Hz countri es or NTSC in 525 line/60 Hz countries. In each case there are many regional variations which require the provision of different equipment. The lack of a single analogue standard also makes it necessary to convert material in many cases, which causes a loss of picture quality.

Figure 1 illustrates the three delivery options used to provide analogue broadcast TV to the majority of viewers in the UK, and they are described below.

- Terrestrial UHF — signals are normally transmitted in a frequency division multiplex (FDM) using vestigial sideband amplitude modulation (VSB/AM) and received using an antenna such as a Yagi array. The UHF receivers are integrated into TV sets. Manufacturers have tried for many years to convince consumers to use the higher quality baseband interface which is now fitted on most new TV sets, satellite set-top boxes and VCRs (referred to as SCART), but they have found in many countries including the UK that they had to include a modulator to convert the baseband signals to UHF, presumably because the facilities provided by these appliances are seen as complementary to the basic terrestrial broadcast services.

- Satellite — signals are normally transmitted in FDM using frequency modulation (FM) with one TV signal occupying a complete satellite transponder (effectively an amplifier and frequency converter) and received using a parabolic dish.

  Significant effort was invested (primarily by manufacturers) in the development of an alternative to PAL/Secam for satellite broadcasting (D/D2-MAC within Europe). This system used multiplexed analogue components (MAC) to improve the picture quality for conventional TV services and to allow for the future introduction of high-definition TV (HDTV) in a compatible manner. It was not adopted because consumers were at that time more interested in paying for the boxes and subscriptions to receive extra TV channels than for increased quality. There was also by then the prospect of digital TV which will offer further increased capacity and flexibility and allow for compatible HDTV.

  Initially, many of the English language satellite TV channels were funded by advertising. Although there are still a few of these, the vast majority are now funded by subscriptions. Over 3.5 million consumers pay for access to the BSkyB services and most of these receive the signals directly from satellite. Conditional access control, based on over-air authorisation of SMART cards in set-top boxes, is used to ensure that consumers pay.

- Cable — signals are normally transmitted in FDM using VSB/AM via a coaxial cable. In most cases a large dish is used to receive satellite services which are then passed on to consumers. In many cases this means that the cable TV services are more expensive than direct satellite reception; however, the cable companies can use different packages of programmes and charging arrangements. Some consumers have found this flexibility plus the avoidance of the need for a dish attractive. Nevertheless, the take-up of cable was relatively slow until the UK regulations were changed to allow cable TV franchise holders to also offer telephony.

Considering the rest of Europe, satellite master antenna (SMATV) networks are also used (particularly in Spain and Italy) to distribute satellite and terrestrial TV signals from a shared dish or dishes throughout blocks of flats using coaxial cables. The signals are normally converted to VSB/AM for transmission, but in some cases the FM satellite format is retained to allow use of standard satellite set-top boxes.

In the USA and Ireland, microwave video distribution (MVDS) is used with VSB/AM as an alternative to cable, once again primarily for distribution of satellite TV services. In this case the signals are transmitted from masts in a similar way to conventional terrestrial TV but at around 2.5 GHz. Modified cable TV set-top boxes are used. BT has also investigated the use of MVDS at 40 GHz with the FM satellite format and a major customer trial was performed at Saxmundham during 1988 [1]. MVDS has also been proposed for use in the West Kent cable TV franchise area.

Figure 1 also shows a telephone which represents the most practical means of interacting with the broadcast service. Telephones and the PSTN are increasingly being used to respond to talent and gameshows and to order goods advertised via the TV channel or associated teletext services. In most cases the interaction is via an operator, but there are also cases where the telephone key pad is used.

A number of cable TV companies in the USA, Canada and Europe are providing interactive services including pay-per-view, teleshopping and games using dedicated low bit rate data modems to respond to a control computer over the coaxial cables.

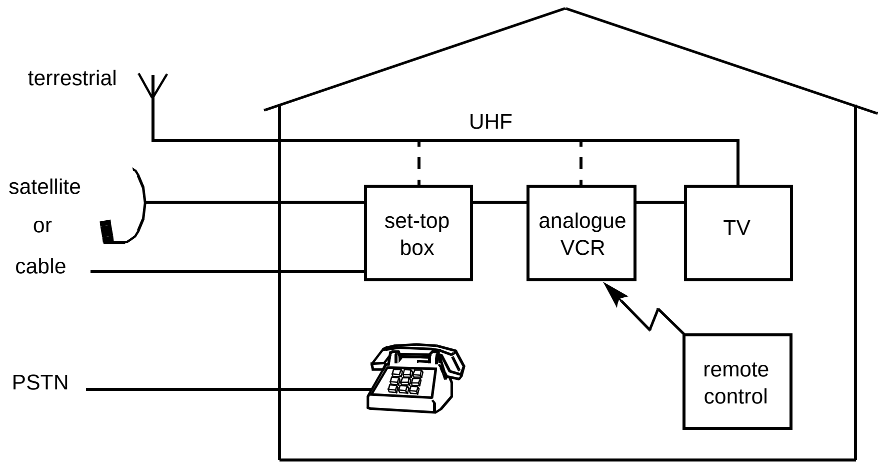

Fig 1 Analogue delivery.

3. Evolution of digital TV standards

In the late eighties, after many years of research into digital TV coding, developments in motion adaptive discrete cosine transform processing and VLSI implementation first made it possible to achieve a picture quality acceptable to most viewers at relatively low bit rates and potential implementation cost. This advance, plus the parallel developments in low bit rate audio coding, provides increased flexibility in terms of the bandwidth required for services plus the benefits of digital coding for transmission and volume production for the first time.

The new digital technology was initially exploited by companies involved with satellite and cable delivery in the USA including General Instruments and Scientific Atlanta. They rapidly developed and implemented proprietary hardware and some of this was put into service. During 1990 General Instruments also proposed the use of digital coding for terrestrial transmission of HDTV within the USA and this was enthusiastically adopted by manufacturers who formed the 'Grand Alliance' to jointly develop a standard.

The development of the early systems was progressed in parallel with the work of the Moving Picture Experts Group (MPEG), often by the same engineers. Many systems were therefore based on the MPEG-1 specification which, although primarily intended for video disk encoding, can also be used at higher rates. RCA (a US subsidiary of Thomson Consumer Electronics) extended this ISO standard and implemented low-cost decoders for the Direct TV satellite broadcasting system. This 150-channel service became operational early in 1994 and has been extremely well received by consumers throughout the USA, even in areas served by cable TV networks.

During 1994 the MPEG-2 system for broadcast TV was also finalised [2—4]. Most companies have now updated their designs to include the enhanced coding and the multiplex formats which it provides and it seems extremely likely that it will steadily become the world standard for digital TV broadcasting. This has allowed the major silicon companies to invest in VLSI chips with confidence and there are already chip sets available. Most companies also have plans for steadily increasing levels of integration which will result in considerable savings on set-top boxes [5]. Services such as Direct TV, which committed to silicon before the finalisation of MPEG-2, will remain incompatible for some time, but some of these companies are now planning to change when further generations of equipment are introduced.

Although the MPEG-2 world standard for coding and multiplexing provides an excellent basis for the introduction of digital TV broadcasting there are, however, many other aspects which have to be addressed and, if possible, standardised. Figure 2 illustrates the most likely scenario for the introduction of digital TV broadcasting.

The same three delivery options are likely to remain dominant within the UK, and in the near term it is likely that dedicated set-top boxes including network-specific receiver/decoders will be provided. Although it would be possible to include more than one receiver/decoder in a box this is not likely to be the norm.

In the short term, it is likely that the output interfaces will remain analogue because consumers will wish to use existing VCRs and TV sets. This could be achieved by PAL encoding and then modulating in VSB/AM on to a UHF carrier, as for current analogue set-top boxes, or by connecting the baseband signals via SCART cables which would preserve the full quality.

The boxes will in many cases include a digital interface to allow data including software and games to be downloaded to local stores in personal computers (PCs) or games machines. This also allows multimedia services

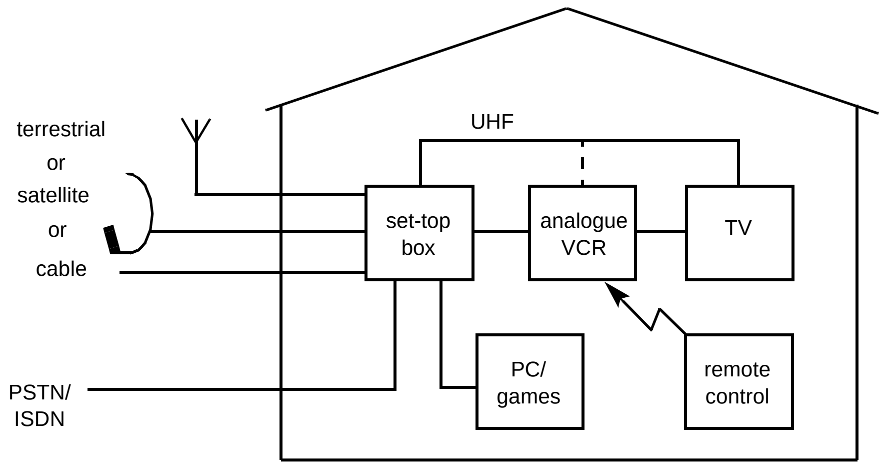

Fig 2 Digital delivery — near term.

including Internet and other specialist news to be downloaded in a cost-effective manner, possibly overnight. This would, however, require wiring between rooms and the effort and cost involved may discourage adoption. The boxes will also, in most cases, have a PSTN (or possibly ISDN) modem incorporated to allow for a range of interactive services associated with broadcasting, as described in section 5. This could in some cases also be used in conjunction with the PC, subject to the same wiring requirements.

Figure 3 shows how the home entertainment environment could develop.

As MPEG-2-based services become established, manufacturers will increasingly incorporate the receiver/ decoders for the most common delivery systems into new TV sets, possibly as plug-in network interface units, purchased and added as required. This could help to sell sets for the wide-aspect ratio HDTV services which are likely to be steadily introduced. The savings in avoiding one or more set-top boxes would help to offset the extra cost of the wide-aspect ratio TV when a new set is required.

These digital systems represent an efficient way of broadcasting popular material. Combining these with the PSTN (or even ISDN) will allow a degree of interactivity and personalisation to be added to broadcasting and its use is expected to build up over time. Ultimately, switched broadband networks could be used to provide broadcast and on-demand services chosen by the individual as an alternative or supplement to other services. Such services are already in the early stages of commercial trials [5].

The other major change is likely to be the implementation and acceptance of a digital alternative to the analogue VCR. Digital VCRs have been specified by a world-wide consensus-forming body termed the DVC

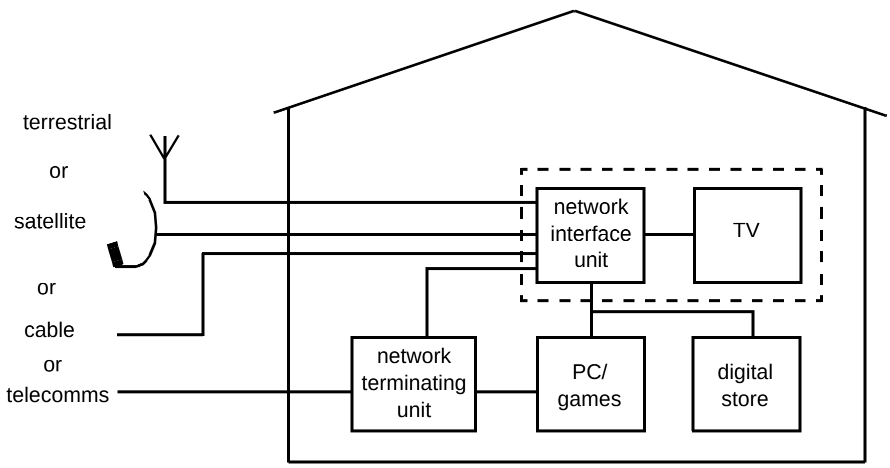

Fig 3 Digital delivery — longer term.

Conference. This specification is intended to allow for analogue inputs and outputs and MPEG coding is not used. It can, however, also be used to record MPEG-encoded signals if a suitable digital interface is provided. The bit rate recorded on the tape is relatively high (25 Mbit/s for conventional TV), partly because the coding is less inefficient than MPEG but also because it is necessary to allow a significant overhead to permit fast forward and other VCR functions to be provided.

A specification has also now been produced for digital VHS recording and playback of MPEG or other data streams and it seems likely that this will allow lower cost implementation for similar functionality. There is also the prospect of recordable disks and even solid-state memories with enough capacity to store a movie. It is very difficult to predict which technology will eventually replace the analogue VCR but it is clear that an interface for a local digital storage capability will be needed.

Two major consensus forming bodies are developing specifications for digital TV broadcasting and the physical and non-physical interfaces required to evolve to the situation illustrated in Fig 3.

3.1 European Digital Video Broadcasting (DVB) project

The European DVB project was officially inaugurated in September 1993 [6, 9]. It grew from the 'European Launching Group for Digital Video Broadcasting' which had already performed a considerable amount of work, particularly on digital terrestrial TV. The project currently consists of a voluntary group of around 180 organisations, including BT, who have signed a 'Memorandum of Understanding' describing the goals of the project. These are to enable the development of standards for digital video broadcasting covering all delivery options and the early introduction of services while avoiding the pitfalls experienced with analogue TV.

The project is funded by the members and has developed its own objectives, policy and rules of procedure, acknowledging that the new digital broadcast and electronic multimedia environment requires market-led approaches to technical development. The members represent four main constituencies, namely consumer electronics manufacturers (including many US and Japanese companies with European subsidiaries), broadcasters and programme providers, network and satellite operators, and regulatory bodies including the European Commission. There is a Steering Board which controls three Commercial Modules dealing with Cable and Satellite, Terrestrial and Interactive services, an *ad hoc* group which has developed a policy for conditional access control and scrambling and the Technical Module which takes 'user requirements' from the commercial modules and generates the specifications to implement them.

The DVB project is not a standards setting body but it has formed a close liaison with the European standards organisations (ETSI and CENELEC) based on a formal co-operation contract. So far five DVB specifications have been passed to ETSI for standardisation and two to CENELEC. The main achievements of the DVB project to date are described in section 4. The decision to add interactivity to the DVB specification was finally made by the Steering Board (following requests from BT and others) during March 1995 [7]. The ongoing and planned work on this topic is described in section 5.

3.2 Digital Audio-Visual Council (DAVIC)

DAVIC is a world-wide consensus-forming body which was formed during 1994 to extend and utilise the work of MPEG. It has around 160 member organisations including large numbers from the USA and Japan where it has a very high profile. The goal is to define specifications for open interfaces and protocols for all audio-visual applications and services based on digital coding and delivery to maximise interoperability across countries and applications/services.

Once again this project is funded by the members who mainly represent manufacturers (mostly computer and cable TV but also consumer electronics) and network operators including telecommunications companies such as BT and a few satellite operators. There are relatively few broadcasters or programme providers and, recognising this, DAVIC is keen to attract extra membership from this community.

DAVIC has a Board of Directors which is advised by a Management Committee and a Strategic Planning Advisory Committee. The Management Committee controls the work of six Technical Committees dealing with Applications, Systems, Technology, Servers, Delivery Systems and Set-Top Units. DAVIC is similar to DVB in that it is not a formal standards organisation. Its aim is to adopt relevant existing standards where possible and, when it produces its own specifications, to submit these to the standards bodies for approval. At the time of writing DAVIC had applied to ISO and the IEC for a formal liaison arrangement similar to that between DVB and ETSI/CENELEC.

The main focus of the first specification (DAVIC 1.0) which is to be finalised in December 1995 is on video-on-demand, but it is also intended to cover interactive broadcast services. The specification differentiates between:

- low-layer protocols and physical interfaces,
- mid-layer protocols,

- high-layer end-to-end protocols.

The physical interfaces include specifications for satellite, asymmetric digital subscriber loop (ADSL), fibre to the kerb, hybrid fibre/coax and ATM delivery to the set-top box which would allow for in-home distribution. An interface is also specified between a network interface unit and a set-top box.

When it was ‘frozen’ at a meeting in June, DAVIC 1.0 included the DVB specification for satellite (with minor optional variations) and cable (after a prolonged debate leading to a vote) except where 256 QAM is used (in some parts of the USA), plus service information (in this case extensions are to be specified to cover NTSC and other missing elements). During the rest of 1995 it is planned to test the current specifications and they should only be changed if the tests prove that they do not fulfil their intended functions.

In the same spirit of co-operation, the work within DVB on interactivity is currently planned to include an analysis of the DAVIC 1.0 specifications, for mid-layer protocols, as discussed in section 5.2. It is likely that they will be used as the basis of the DVB specification, providing they can be implemented at a low enough cost for PSTN/ISDN-based interactive applications.

4. Digital video broadcasting standards

The DVB specifications include elements which are applicable in all cases and others covering modulation and coding for the different delivery systems.

4.1 Common elements

The aim of these specifications is to maximise commonality so that manufacturers can reuse as many elements as possible across the complete range of delivery systems.

MPEG architecture

The MPEG-2 standard [2—4] includes a family of video coding standards [8, 9]. These are separated into levels which define the number of pixels and profiles which specify the processing used. DVB has selected the Main Level and Main Profile option which has 720 $\times$ 576 pixels with B-frame coding because it most closely matches the anticipated service requirements. It has also selected a subset from the remaining toolkit and this is described in a guidelines document (one of a set of DVB blue books) with which manufacturers will voluntarily comply in their development and implementation of set-top boxes and encoders. The remaining flexibility in terms of the coding and resolution still allows a number of options to be implemented ranging from limited definition TV (LDTV) at bit rates as low as 1.5 Mbit/s to enhanced definition TV (EDTV) with wide aspect ratio at up to 15 Mbit/s. Audio coding is according to the MPEG layer 2 (MUSICAM) specification.

Service information

DVB has defined a set of service information (SI) tables which are transmitted with broadcast services to assist with the implementation of user interfaces and this specification has now been standardised by ETSI and adopted by DAVIC [10]. The SI tables contain information about each specific programme, packages of programmes from a specific broadcaster, the current event within the programme and other information. This will enable a viewer to navigate through the many channels which will become available from different sources not only on a channel-by-channel basis but also by selecting categories of services. Information will also be supplied to allow parents to decide on suitability of programming for children and for electronic programme guides and many other desirable new features.

Teletext

A specification for carriage of conventional teletext has been standardised by ETSI [11]. In practice, many broadcasters are likely to introduce new graphics-based equivalents.

Subtitling

A specification for subtitling including the capability to deliver station logos and bitmaps for other graphics applications is currently being finalised and it will be passed to ETSI for standardisation in the near future.

Conditional access

Conditional access (CA) control is essential for broadcast subscription TV as it ensures that consumers pay for the services. In most cases, consumers are supplied with a SMART card when they subscribe to the services and this is then plugged into a CA module inside the set-top box via a hole in the case and authorised via the broadcast channel. The CA module allows for the secure distribution of decryption keys which can be used to descramble services. This is a commercially sensitive subject and DVB has not attempted to define a standard CA system. It has, however, defined a common scrambling algorithm and fully specified an interface allowing boxes to accommodate more than one PCMCIA mounted CA module [12]. This specification is currently being considered for standardisation by CENELEC. It has also been submitted to DAVIC.

Physical interfaces

DVB has defined a set of physical interfaces covering inputs from all of the different network options and outputs to the TV set and to other peripherals and this has been submitted to CENELEC for standardisation [13]. The interfaces are optional, but, if an interface is used, then it should comply with the DVB requirements to ensure commonality. An RS232 data port has been defined for communication with a PC or other device.

A integral PSTN modem is specified to allow for the interactive services described in section 5. One or more of the following modes should be supported — automode selection ITU V.21, V.23 (1200/75), V.22 or V.22 bis transmission protocols. It is also recommended that V.32, V.32 bis and V.34 support should be included and that designs should not preclude the addition of future enhancements. A connector allowing for an external PSTN or cable TV modem is specified. Control is via the Hayes AT command set.

The interfaces required for digital video cassettes (DVCs) have also been investigated, as discussed in section 3. Initially, a number of serial data interfaces were considered, but, following reports of support from the DVC Conference for the IEEE P1394 specification, it was decided to concentrate on this. The ongoing work is concerned with defining how a DVB MPEG transport stream can be carried over this interface in the most efficient manner. It is likely that the work will be extended to allow for the other digital storage systems discussed in section 3, and once again the choice of a physical interface will be influenced by the other bodies defining these systems.

Although the IEEE P1394 specification is suitable for connections between clusters of equipment close to each other it requires repeaters every 4.6 m which makes it impractical for home distribution applications including connection of telecommunications services from a network terminating unit to a network interface unit, as shown in Fig 3. DVB has therefore also investigated the requirements for a longer reach data interface. Although a number of suitable candidates were identified it was decided to await developments in the relevant standards bodies before making a selection. It is worth noting that DAVIC has in the meantime selected a 25-Mbit/s ATM-based system which could be suitable for the home distribution applications.

4.2 Delivery systems

These specifications define the modulation and coding for all of the delivery options considered by DVB so far. The Reed Solomon outer forward error correction (FEC) coding, which is used to combat the effects of impulsive noise, is the same for all delivery options. A common concatenated inner FEC code is also used for terrestrial and satellite delivery. This eases the design of the VLSI chips and reduces the complexity of transconversion between the satellite and cable formats. In all cases a single MPEG transport stream is used to carry a flexible mixture of video audio and data channels in a time division multiplex (TDM). This means that the interface between the receiver/decoder (or network interface unit) and the MPEG decoder/ demultiplexer can be the same for all delivery options.

Terrestrial

Although digital terrestrial TV (DTT) was the first delivery system considered by DVB, the specification is not yet approved for standardisation. The reason is that there is very little spectrum available for new digital services in many countries and it is therefore necessary to allow for a range of applications and introduction scenarios. The BBC has, however, now decided to proceed with trials of DTT which should lead to services during 1997. The Government has now also issued a White Paper with proposals for DTT introduction, using allocations which are not used for analogue services to avoid unacceptable interference. The digital signals cause less interference and are affected less by interference. It is predicted that 18 new TV channels could be introduced over most of the UK. This assumes that each DTT carrier is used for an MPEG transport stream at around 18—20 Mbit/s shared between three TV channels in TDM. This has given an increased impetus to the work and it is likely that the specification will be approved by DVB during December 1995.

The current specification is based on coded orthogonal frequency division multiplexing (COFDM) with associated convolutional inner FEC coding. Three modulation options are available and the inner FEC coding can operate at a range of rates. The symbol rate available for the MPEG transport stream can also be varied. This makes it possible to trade off capacity against performance in a flexible manner (from around 6 to 32 Mbit/s depending on carrier-to-noise requirements), allowing the same system to be used for different coverage and interference scenarios.

The COFDM process requires significant complexity, and this means that the receiver/decoders will be more expensive compared with those for satellite and cable. Nevertheless, the advantage of reception via existing (and well-accepted) Yagi arrays and the attraction of an enhanced ‘free to air’ service (licence and advertising funded), plus new subscription services (probably using the wide-aspect ratio format from the start) are likely to ensure the volumes required to rapidly reduce the costs.

Satellite

The DVB satellite TV modulation and coding specification has now been standardised by ETSI and adopted by DAVIC [14]. The system includes flexibility in terms of the inner FEC coding rate and the symbol rate to allow for a range of satellite transponder bandwidths (26 MHz to 72 MHz) and powers (49 dBW to 61 dBW). QPSK modulation is used for transmission [15].

A number of satellite operators are planning to provide capacity for direct-to-home services and these will mainly have bandwidths of around 33 MHz and power of around 52 dBW, allowing around 39 Mbit/s to be delivered via a 60 cm dish with an acceptable margin. A 45 cm squarial antenna could be used if a lower margin was accepted or if the data rate was reduced. It is worth noting that with a digital system the picture quality normally remains constant until the carrier level becomes too low, for instance during a rain fade, when it can fail completely. It is therefore necessary to use a higher margin than for analogue satellite TV which degrades more gracefully.

BSkyB have transmitted signals via the Astra 1D satellite for demonstration purposes and to allow set-top box manufacturers to ensure that their designs are compliant. Other similar test transmissions are likely and services are likely to start soon. BSkyB is reported to be planning a 120-channel English language service via the Astra 1E satellite and further services via the Eutelsat ‘Hotbird 3’ for other European markets. EchoStar is also planning a 100-channel service in the USA for mid-1996. In fact, services are likely to start in France, South Africa and the Far East before this.

Cable

The DVB cable TV modulation and coding specification has also now been standardised by ETSI and adopted by DAVIC [16]. It was completed after the satellite specification in order to ensure maximum commonality for simple transmultiplexing. The signals are transmitted using quadrature amplitude modulation (QAM). The normal mode is likely to be 64 QAM but 16 QAM and 32 QAM are also specified and these may be used for SMATV, as described below. Higher rate systems are also defined, but their use will depend on the capacity of the cable network to deal with the reduced data eye height (and the fact that DAVIC has defined a different system for these rates).

It is possible to vary the symbol rate (the maximum rate will be around 39 Mbit/s for 64 QAM) as for satellite, and this makes it possible to assemble multiplexes including TV channels from many sources with a different conditional access control system to that used for satellite delivery. This implies significant extra complexity and it is likely that in some cases the cable operators will come to an arrangement with the satellite broadcasters over use of their conditional access control systems. The main markets for digital cable TV in Europe are currently seen to be in France and Germany.

SMATV

The DVB satellite master antenna TV (SMATV) specification which has also now been standardised by ETSI, includes two options [17]. For the first, the satellite signals are transmitted via coaxial cables at the intermediate frequency. This allows the use of standard DVB satellite TV set-top boxes. In the second case, the signals are trans-multiplexed to the cable TV format allowing the use of standard DVB cable set-top boxes. The equalisation requirements for the cables are defined and for new installations it should be possible to use the 64 QAM cable TV mode. However, for older installations the lower-rate options may have to be used. This would imply increased complexity and cost, not least because the existing amplifiers would have to be replaced, and so some networks may stay with analogue until they are upgraded.

MVDS

Work to define a specification for microwave video distribution has recently started. This approach is seen as a cost-effective way to increase the reach of cable TV networks in rural areas. It is expected that there will be two options based on the existing satellite and cable specifications and boxes, as for SMATV.

5. Addition of network interactivity

The Interactive Services Commercial Module (ISCM) was set up following the DVB decision to include interactivity in its specifications [7]. This module is jointly chaired by BT and Betatechnic from Germany. Two *ad hoc* groups have also been set up within the Technical Module to define:

- modulation and coding for the complete range of return channels associated with the delivery systems,
- protocols for interactive services (ideally independently of the delivery system) to implement the user requirements defined by the ISCM.

The first *ad hoc* group is jointly chaired by Hispasat of Spain and the ITC and the second is chaired by BT.

5.1 User requirements

The ISCM is defining user requirements for asymmetric interactive services supporting broadcast to the home with narrowband return channels comparable with those offered by PSTN/ISDN, as a first priority. Four different types of interaction (in addition to local interaction with downloaded software) can be identified, each with increasing levels of extra functionality. It is still to be decided which of the necessary functions DVB will specify.

Response to broadcast service

In this case the user can respond to broadcast material in an anonymous manner, for instance for voting in a talent show or responding to an opinion poll. It is only necessary to count the number of votes. There is a problem with this type of service in that very large numbers of short calls can result. In some cases it would be necessary to randomise the responses in time to spread the traffic peaks or even to use store and forward. This type of interaction should therefore ideally be controlled as part of the TV programme (perhaps with the process explained verbally by the presenter). This also helps with another problem which can arise if on-screen graphics are overlayed on to an existing programme with icons, subtitles or other graphics which would be obscured.

Information exchange

The simplest example of this type of service is where a response to a gameshow involves a prize. This makes it necessary for the participant to be identified and perhaps for an acknowledgement to be sent. In this case a one-to-one dialogue is required, plus some form of on-screen display. The same facilities can also be used to request personalised information and to deliver it to the set-top box. A user could for instance request information from a remote database with details related to a sporting event or other ongoing broadcast service. The information could be textual but it would also be possible to use MPEG-2 video coding to deliver still pictures which could be decoded by the processing already in the set-top box. Alternatively, the audio processing and interfaces in the box could be used to implement an audio response facility. The conditional access control capabilities in the box could also be exploited to add security.

Purchase requests

In this case, consumers can purchase access to broadcast services possibly in an impulse pay-per-view mode. This would probably be implemented as an extension to the broadcast conditional access control system. Purchase of other goods and services associated with teleshopping, telebanking and gambling would involve more generalised financial transactions which implies the addition of an independent financial security system to confirm the identity of the user rather than the user's equipment and to confirm financial status. The delivery could be over-air in some cases (as for instance with software) or by conventional mail, and store and forward would probably be acceptable.

Messaging and remote monitoring

This case covers three types of service. Firstly, the user can initiate a message to the service or network provider, for instance requesting help. Secondly, the service provider can send messages to users, or one user in particular, perhaps to inform them of new software releases, or to the network provider to flag that the return channel is not working. Thirdly, the network provider can send messages to the user's box requesting it to send, for example, viewing statistics or bit error rate over the last 24 hours at a particular time, or to the service provider to inform him that particular measures have been taken to avoid network congestion. An alternative would be for users to programme their home video recorder to a particular setting from the office.

5.2 Plans for implementation

Until the user requirements for interactive services are completed and formally approved it is not possible to say with certainty what will be specified by DVB. Nevertheless, it is possible to describe the work which is proposed in general terms.

There is a clear requirement to specify other return channel options in addition to the PSTN. A number of cable operators are particularly keen to progress this and it is likely that this will be given a high priority by the new Technical Module *ad hoc* group which is being set up.

The Systems for Interactive Services *ad hoc* group tasked with specifying the required protocols for interactive services was formed from an existing group and it has therefore already started work. A specification was produced by this group for a telephone descriptor and this forms part of the DVB service information (SI) standard described in section 4.1. This descriptor can be used to deliver telephone numbering information to a set-top box allowing it to dial automatically when the user instructs it to do so via his remote control. It is already recognised that further information will have to be delivered over the broadcast channel and it is intended to use this descriptor, plus other parts of the SI specification, as the basis. The group has identified three tasks.

Analyse the user requirements

It is necessary to analyse the output of the commercial module as it develops and to identify the implications for standards, bearing in mind what already exists within DAVIC and other bodies, so that DVB can agree on which of the required functions it intends to specify.

Analyse the DAVIC specification

Ideally the DAVIC specification should meet many of DVB's requirements for interactive TV protocols. It is therefore intended to investigate the implementation implications of the existing specifications and identify any requirements which are not currently covered.

Extend the existing specifications

If there are requirements which cannot be met with existing specifications, extensions or new specifications will be developed, ideally with the knowledge and support of DAVIC and any other relevant bodies. The extensions to the telephone descriptor which have already been identified are likely to fall into this category. As they are based on an existing DVB standard which has been adopted by DAVIC, it is logical for DVB to lead on this work.

6. Conclusions

Broadcast technology is developing rapidly and the change from analogue to digital is almost upon us. It is likely that satellite services will be introduced first within Europe, but terrestrial digital TV services will follow rapidly, at least within the UK. The terrestrial services have the advantage that a dish is not required and that an enhanced package of 'free to air' services are likely to be provided in addition to new subscription services. In both cases it is likely that network interaction via the PSTN (or possibly the ISDN) will be offered as an integral part of the new services.

Specifications for the core processing and delivery of the broadcast services have been produced by the European DVB project and these are being standardised. Work within DVB has now started on specifications allowing the addition of interactivity. This work is at an early stage and it is expected that the specifications produced by the DAVIC project will be used as the basis of the DVB specification, providing the cost of the hardware and software to implement interactivity is acceptable. The work within DVB will concentrate on the requirements for interactive services to complement broadcasting, but ideally also allow for independent network-based services.

Appendix

Glossary

ADSL	asymmetric digital subscriber loop
ATM	asynchronous transfer mode
CENELEC	European Committee for Electrotechnical Standardisation
COFDM	coded orthogonal frequency division multiplex
DAVIC	Digital Audio-Visual Council
DVB	European Digital Videobroadcasting project
DVC	digital video cassette
ETSI	European Telecommunications Standards Institute
FDM	frequency division multiplex
FM	frequency modulation
IEC	International Electrotechnical Commission
ISO	International Organisation for Standardisation
ITU	International Telecommunications Union
MVDS	microwave video distribution system
PCMCIA	Personal Computer Memory Card International Association
QAM	quadrature amplitude modulation
QPSK	quaternary phase shift keying
SMATV	satellite master antenna TV
VSB/AM	vestigial sideband amplitude modulation

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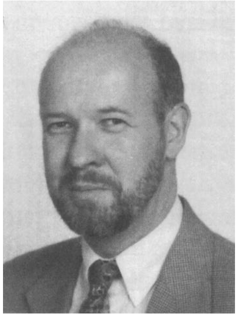

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